

Why Transformer Attention is Low-Rank

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LayerNorm subtracts the mean. That's it.

The Claim

Transformers apply LayerNorm before attention. LayerNorm subtracts the mean, forcing all vectors onto a $(d-1)$ -dimensional hyperplane. Since Q and K are linear projections of these vectors, $\text{rank}(QK^\top) \leq d-1$.

The Proof (2D $\rightarrow$ 1D)

Take $\mathbf{x} = [x_1, x_2]$. Subtract the mean $\mu = (x_1 + x_2)/2$. The result $\mathbf{z} = [x_1 - \mu, x_2 - \mu]$ satisfies:

$$z_1 + z_2 = (x_1 - \mu) + (x_2 - \mu) = x_1 + x_2 - 2\mu = 0$$

So $z_2 = -z_1$. Every centered vector is $\mathbf{z} = [a, -a] = a \cdot [1, -1]$. Stack N such vectors:

$$\begin{bmatrix} 1.2 & -1.2 \\ -0.8 & 0.8 \\ 2.0 & -2.0 \\ 0.5 & -0.5 \end{bmatrix} = \begin{bmatrix} 1.2 \\ -0.8 \\ 2.0 \\ 0.5 \end{bmatrix} \times \begin{bmatrix} 1 & -1 \end{bmatrix}$$

Outer product: $(4 \times 1) \times (1 \times 2) \rightarrow (4 \times 2)$ matrix. **Rank = 1**, no matter how many rows.

Generalization to d Dimensions

In d dimensions: $z_1 + z_2 + \dots + z_d = 0$. One constraint on d variables $\rightarrow (d-1)$ -dimensional subspace. The $[1, 1, \dots, 1]$ direction is gone. Any matrix of such vectors has $\text{rank} \leq d-1$. Linear projections $Q = W_q \mathbf{z}$ and $K = W_k \mathbf{z}$ cannot increase rank, so $\text{rank}(QK^\top) \leq d-1$.

Verification

Any vector decomposes as: $\mathbf{x} = \mu \cdot [1, 1, \dots, 1] + (\mathbf{x} - \mu)$

First term: rank-1. Second term: LayerNorm output ($d-1$ dimensions). Adding one rank-1 component reconstructs the original. That component is exactly what LayerNorm deletes. ■